卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-EC-0201 30-B003701S-D0201

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

电气 专业 第 02 卷 第 01 册
Electrical SPECIALITY VOLUME 02 BOOK 01

卷册名称 11kV站用电系统及设备安装
Volume name 11kV Station Power System and Equipment Installation

批　准 APPROVED BY	2019.06.10
审　核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	魏翀 2019.06.10
图纸 7 张 DRAWING 7 NOS.	打印件 本 PRINTED COPIES NOS.

序号 No.	图　号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-D0201-001	卷册说明 Volume instructions and legend	A	1		
2	30-B003701S-D0201-002	11kV站用电系统接线示意图 Single Line Diagram of 11kV Station Power System	A	1		
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5	30-B003701S-D0201-005	11kV开关柜正视图 Front view of 11kV cabinets	A	1		
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注:

卷册说明
Volume instructions and legend

一、项目概况：
First: Project overview
项目名称：巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程/拉合尔±660kV换流站
Project Name: Pakistan Mertiary-Lahore +660 kV HVDC Project/Lahore +660 kV Converter Station
建设单位：国家电网/中国电力技术装备有限公司
Development organization: State Grid / China Electric Power Equipment And Technology CO., LTD
本次出图范围为:11kV站用电系统及设备安装
Design scope: 11kV Station Power System and Equipment Installation
二、设计内容
Second: Design content

1.本卷册包括全站11kV站用电系统及其材料，包括35kV和132kV站用变压器低压侧户外11kV电缆终端。11kV干式变高压侧的11kV户内电缆终端开列在本卷册中。

1. This volume includes the entire 11kV station power system and its materials, including 11kV outdoor cable terminal on low-voltage side of 35kV and 132kV station transformer. 11kV indoor cable terminals on the high-voltage side of 11kV dry transformer are listed in this volume.

2.11kV干式变压器的安装不属于本卷册设计范围，详见D0204卷册。

2.Installation of 11 kV dry-type transformer does not belong to the scope of this volume design. See volume D0204 for details.

三、11kV供配电系统
Third: 11kV Augxilliary system

1、全站设2台500/35kV站用降压变压器，高压侧分别接入两组交流500kV母线，低压侧接35/11kV站用变压器，通过35/11kV站用变压器降压后经11kV电缆接入站11kV系统；站用备用电源由附近132kV线路T接经132kV架空线路接入站内，通过1台132/11kV站用变压器降压后经11kV电缆接入站11kV系统。

1.The project include 2 500/35kV transformers which the power from 500kV busbar and low voltage terminal is linked to 35/11kV auxiliary transformers.And the power is send to 11kV system by 35/11kV auxiliary transformers.In addition,there is a outcoming power line from existed 132kV transmission line outside the converter station,stepped down by 132/11kV transformer and connected to 11kV system.

2、站11kV系统接线采用单母线，设2个工作段和1个备用段，每段各由一回独立的电源供电。每个工作段均可从备用母线段取得备用电源。

2.The 11kV single bus include 2 working sections and a backup section which is supplied by an independent power source.Each working section could get power from the backup when malfunction happens.

四、开关柜安装位置和电缆供货范围
Fourth.Installation location and supply scope

1、本工程11kV开关柜采用11kV金属铠装移开式开关柜。11kV开关柜体均安装于11kV配电室，11kV配电室与500kV第一继电小室和计量室合建。

1.The 11kV metalized armor traveling cabinets are installed in 11kV station service power room which is built with 500kV No.1 relay house and Metering Room together.

五、其他

1、本站所有11kV电缆布置在电缆沟或埋管内。

1.All 11kV cables in this station are arranged in cable trenches or buried pipes.

2、所有11kV电缆应参照路径图进行敷设，电缆转弯半径一般不得小于15d（d为电缆直径），具体以厂家要求为准。3回11kV进线电源电缆分沟敷设。从不同段11kV母线引接至各动力中心的电缆应尽量分沟敷设，当只能敷设在同一条电缆沟中时，应敷设在不同侧的电缆支架上。

2.All 11kV cables should be laid according to the road map. The turning radius of the cables should not be less than 15d (d is the diameter of the cables). Specific requirements of the manufacturer shall prevail. Three 11 kV power supply cables should be laid in differernt trenches。Cables from different sections of 11kV buses should be laid in differernt trenches if possible. When they can only be laid in the same trench, they should be laid on cable brackets on different sides.

3、应按电压等级由高至低的电力电缆“由下而上”的顺序敷设；其中，最大截面电缆宜敷设在电缆沟最下层支架。11kV 电缆应与400V动力电缆、控制电缆之间采用防火隔板进行分隔。

3.The cable should be laid in the order of "bottom-up" from the highest to the lowest voltage level, and the maximum cross-section cable should be laid on the lowest bracket of the cable trench.11kV cable should be separated from 400V power cable and control cable by fire-proof partition board.

4、交流电力电缆的铠装层应可靠接地。三芯电缆的金属层应在电缆线路两终端分别接地。

4. The armor layer of AC power cable should be grounded reliably. The metal layer of the three-core cable should be grounded separately at the two terminals of the cable line.

5、电缆敷设须满足GB50217-2017电力工程电缆设计规范。

5.Cable laying should meet GB50217-2017 Electric Power Engineering Cable Design Specification

6、其他电缆敷设事宜可参见本工程D0406、D0407册。

6.For other cable laying matters, please refer to the volume D0406、D0407 of this project.

Reference Drawings

Drwg. No.	Title

Revisions	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR
国家电网 STATE GRID
中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR
East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

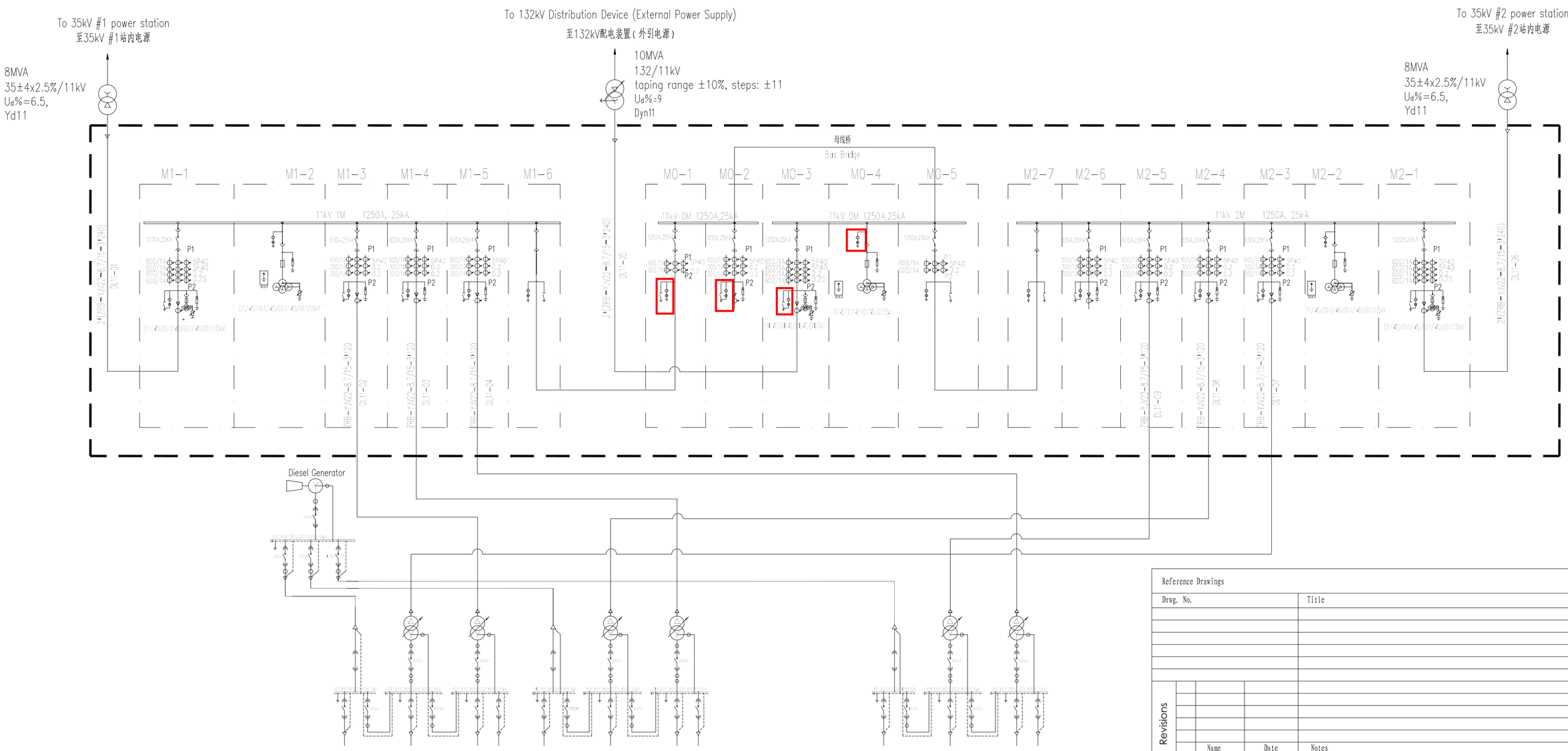
MLHVDC-LCS-ECI-DD-EC-0201-001A

	姓名 Name	日期 Date
设计 Designed	魏聊	2019.06.07
校核 Checked	郑煜	2019.06.08
审核 Reviewed	张谢平	2019.06.09
批准 Approved	潘益竿	2019.06.10

卷册说明
Volume instructions and legend

比例 Scale:
图号 DWG NO: 30-B003701S-D0201-001
版本 REV: A

A3(420x297)



注:
1、虚线框内为本卷册设计范围。

NOTE:
1. The design scope of this volume is in the dotted frame.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: Detail Design

MLHVDC-LCS-ECI-DD-EC-0201-002A

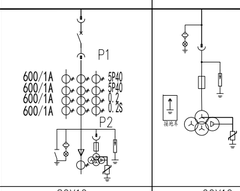
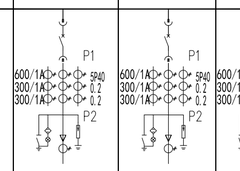
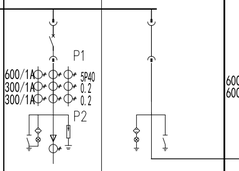
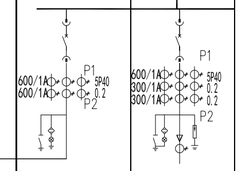
	姓名 Name	日期 Date
设计 Designed	魏翀	2019.06.07
校核 Checked	郑旭	2019.06.08
审核 Reviewed	张谢平	2019.06.09
批准 Approved	潘益平	2019.06.10
比例 Scale		图号 DWG NO. 30-B003701S-D0202-002

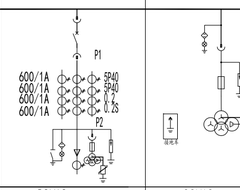
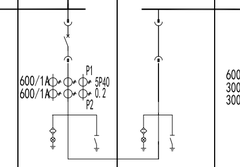
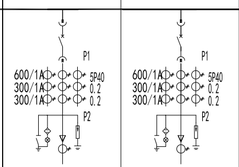
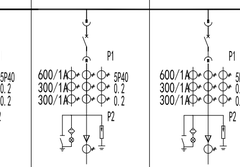
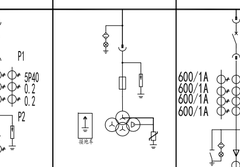
11kV站用电系统接线示意图

Single Line Diagram of 11kV Station Power System

版本 REV. A

A2(594x420)

Cabinet type: KYN28-12A	柜型:KYN28-12A								母线桥连接至M0-5 母线桥
Rated / operating voltage: 12/11kV	额定/运行电压:12/11kV								
Main bus current: 1250A	主母线电流:1250A								
Main bus type: TMY-80X10	主母线类型:TMY-80X10								
Power frequency / impact withstand voltage: 42/75kV	工频/冲击耐压:42/75kV								
Control power supply: DC220V	控制电源:DC220V								
Energy storage power supply: DC220V	储能电源:DC220V								
Protection level: IP4X	防护等级:IP4X								
One-plan one-line diagram	一次方案单线图								
Branch busbar specification TMY	分支母排规格TMY	80X10	60X10	60X10	60X10	60X10	80X10	80X10	60X10
Switch cabinet phase sequence	开关柜相序								
use	用途	进线柜	母设柜	馈线柜	馈线柜	馈线柜	隔离柜	分段柜	馈线柜
Cabinet number	柜号	M1-1	M1-2	M1-3	M1-4	M1-5	M1-6	M0-1	M0-2
Main switch	主开关	VD4/P,HE,1250A,25KA	熔断器+避雷器手车	VD4/P,HE,630A,25KA	VD4/P,HE,630A,25KA	VD4/P,HE,630A,25KA	隔离手车1250A	VD4/P,HE,1250A,25KA	VD4/P,HE,630A,25KA
Grounding switch / isolating switch	接地开关/隔离开关	JN15-12		JN15-12	JN15-12	JN15-12	JN15-12	JN15-12	JN15-12
control method	控制方式	LOCAL/REMOTE		LOCAL/REMOTE	LOCAL/REMOTE	LOCAL/REMOTE		LOCAL/REMOTE	LOCAL/REMOTE
High voltage fuse	高压熔断器	XRNP-12/0.5	XRNP-12/0.5						
Fault indicator	故障指示器								
Current Transformer	电流互感器	3*LZZBJ9-600/1A, 5P40/5P40/0.2/0.2S ,15/15/10/10VA		3*LZZBJ9 600/1A,5P40,20VA,300/1A 0.2/0.2,10VA/10VA	3*LZZBJ9 600/1A,5P40,20VA,300/1A 0.2/0.2,10VA/10VA	3*LZZBJ9 600/1A,5P40,20VA,300/1A 0.2/0.2,10VA/10VA		3*LZZBJ9 600/1A 5P40/0.2,20/10VA	3*LZZBJ9 600/1A,5P40,20VA,300/1A 0.2/0.2,10VA/10VA
Zero sequence current transformer	零序电流互感器	Φ240,100/1A,10P10,5VA		Φ120,100/1A,10P10,2.5VA	Φ120,100/1A,10P10,2.5VA	Φ120,100/1A,10P10,2.5VA			Φ120,100/1A,10P10,2.5VA
Voltage transformer / transformer	电压互感器/变压器	3*JDZX11-10 (11/√3)/(0.1/√3)/(0.1/√3)/(0.1/3)kV 20VA/20VA/100VA 0.2(3P)/0.2(3P)/3P	3*JDZX11-10 (11/√3)/(0.1/√3)/(0.1/√3)/(0.1/3)kV 20VA/20VA/100VA 0.2(3P)/0.2(3P)/3P						
Arrester/overvoltage protector	避雷器/过电压保护器	YH5WZ2-17/45Q	YH5WZ2-17/45	YH5WZ2-17/45	YH5WZ2-17/45	YH5WZ2-17/45			YH5WZ2-17/45
Relay protection	继电保护	PCS-9611-EN		PCS-9671-EN	PCS-9671-EN	PCS-9671-EN		PCS-9611-EN	PCS-9671-EN
Multi-function meter	多功能电度表								
Ammeter / voltmeter	电流表/电压表	3*A,1*V	1*V	3*A	3*A	3*A		3*A	3*A
Microcomputer harmonic elimination device	微机消谐装置	WNXIII-10/A	WNXIII-10/A						
Note 1: Intelligent control device	备注1: 智能操控装置	HZYN-5800H	HZYN-5800H	HZYN-5800H	HZYN-5800H	HZYN-5800H	HZYN-5800H	HZYN-5800H	HZYN-5800H
Note 2: Electromagnetic lock	备注2: 电磁锁		DSN-BMY						
Note 3: One harmonic elimination device	备注3: 一次消谐器	LXQ(D)II-12	LXQ(D)II-12						
Note 4:	备注4:								
Primary cable cross section (mm²)	一次电缆截面(mm²)	2*(ZRB-YJV22-8.7/15-3*240)		ZRB-YJV22-8.7/15-3*120	ZRB-YJV22-8.7/15-3*120	ZRB-YJV22-8.7/15-3*120			
Cabinet size (width * depth * height) mm	柜体尺寸(宽*深*高)mm	800*1513*2320	800*1513*2320	800*1513*2320	800*1513*2320	800*1513*2320	800*1513*2320	800*1513*2320	800*1513*2320
Quadratic map number	二次图图号								

Cabinet type: KYN28-12A	柜型:KYN28-12A			母线桥连接至M0-2 母线桥							
Rated / operating voltage: 12/11kV	额定/运行电压:12/11kV										
Main bus current: 1250A	主母线电流:1250A										
Main bus type: TMY-80X10	主母线类型:TMY-80X10										
Power frequency / impact withstand voltage: 42/75kV	工频/冲击耐压:42/75kV										
Control power supply: DC220V	控制电源:DC220V										
Energy storage power supply: DC220V	储能电源:DC220V										
Protection level: IP4X	防护等级:IP4X										
One-plan one-line diagram	一次方案单线图										
Branch busbar specification TMY	分支母排规格TMY	80X10	60X10	80X10	80X10	60X10	60X10	60X10	60X10	60X10	80X10
Switch cabinet phase sequence	开关柜相序										
	用途	进线柜	母设柜	分段柜	隔离柜	馈线柜	馈线柜	馈线柜	馈线柜	母设柜	进线柜
Cabinet number	柜号	M0-3	M0-4	M0-5	M2-7	M2-6	M2-5	M2-4	M2-3	M2-2	M2-1
Main switch	主开关	VD4/P,HE,1250A,25KA	熔断器+避雷器手车	VD4/P,HE,1250A,25KA	隔离手车1250A	VD4/P,HE,630A,25KA	VD4/P,HE,630A,25KA	VD4/P,HE,630A,25KA	VD4/P,HE,630A,25KA	熔断器+避雷器手车	VD4/P,HE,1250A,25KA
Grounding switch / isolating switch	接地开关/隔离开关	JN15-12		JN15-12	JN15-12	JN15-12	JN15-12	JN15-12	JN15-12		JN15-12
control method	控制方式	LOCAL/REMOTE		LOCAL/REMOTE		LOCAL/REMOTE	LOCAL/REMOTE	LOCAL/REMOTE	LOCAL/REMOTE		LOCAL/REMOTE
High voltage fuse	高压熔断器	XRNP-12/0.5	XRNP-12/0.5							XRNP-12/0.5	XRNP-12/0.5
Fault indicator	故障指示器										
Current Transformer	电流互感器	3*LZZBJ9-600/1A, 5P40/5P40/0.2/0.2S 15/15/10/10VA		3*LZZBJ9 600/1A 5P40/0.2,20/10VA		3*LZZBJ9 600/1A,5P40,20VA,300/1A 0.2/0.2,10VA/10VA	3*LZZBJ9 600/1A,5P40,20VA,300/1A 0.2/0.2,10VA/10VA	3*LZZBJ9 600/1A,5P40,20VA,300/1A 0.2/0.2,10VA/10VA	3*LZZBJ9 600/1A,5P40,20VA,300/1A 0.2/0.2,10VA/10VA		3*LZZBJ9-600/1A, 5P40/5P40/0.2/0.2S ,15/15/10/10VA
Zero sequence current transformer	零序电流互感器	Φ240,100/1A,10P10,5VA				Φ120,100/1A,10P10,2.5VA	Φ120,100/1A,10P10,2.5VA	Φ120,100/1A,10P10,2.5VA	Φ120,100/1A,10P10,2.5VA		Φ240,100/1A,10P10,5VA
Voltage transformer / transformer	电压互感器/变压器	3*JDZX11-10 (11/√3)/(0.1/√3)/(0.1/√3)/(0.1/3)kV 20VA/20VA/100VA 0.2(3P)/0.2(3P)/3P	3*JDZX11-10 (11/√3)/(0.1/√3)/(0.1/√3)/(0.1/3)kV 20VA/20VA/100VA 0.2(3P)/0.2(3P)/3P							3*JDZX11-10 (11/√3)/(0.1/√3)/(0.1/√3)/(0.1/3)kV 20VA/20VA/100VA 0.2(3P)/0.2(3P)/3P	3*JDZX11-10 (11/√3)/(0.1/√3)/(0.1/√3)/(0.1/3)kV 20VA/20VA/100VA 0.2(3P)/0.2(3P)/3P
Arrester/overvoltage protector	避雷器/过电压保护器	YH5WZ2-17/45Q	YH5WZ2-17/45			YH5WZ2-17/45	YH5WZ2-17/45	YH5WZ2-17/45	YH5WZ2-17/45	YH5WZ2-17/45	YH5WZ2-17/45Q
Relay protection	继电保护	PCS-9611-EN		PCS-9611-EN		PCS-9671-EN	PCS-9671-EN	PCS-9671-EN	PCS-9671-EN		PCS-9611-EN
Multi-function meter	多功能电度表										
Ammeter / voltmeter	电流表/多功能表	3*A,1*V	1*V	3*A		3*A	3*A	3*A	3*A	1*V	3*A,1*V
Microcomputer harmonic elimination device	微机消谐装置	WNXIII-10/A	WNXIII-10/A							WNXIII-10/A	WNXIII-10/A
Note 1: Intelligent control device	备注1: 智能操控装置	HZYN-5800H	HZYN-5800H	HZYN-5800H	HZYN-5800H	HZYN-5800H	HZYN-5800H	HZYN-5800H	HZYN-5800H	HZYN-5800H	HZYN-5800H
Note 2: Electromagnetic lock	备注2: 电磁锁		DSN-BMY							DSN-BMY	
Note 3: One harmonic elimination device	备注3: 一次消谐器	LXQ(D)II-12	LXQ(D)II-12							LXQ(D)II-12	LXQ(D)II-12
Note 4:	备注4:										
Primary cable cross section (mm²)	一次电缆截面(mm²)	2*(ZRB-YJV22-8.7/15-3*240)					ZRB-YJV22-8.7/15-3*120	ZRB-YJV22-8.7/15-3*120	ZRB-YJV22-8.7/15-3*120		2*(ZRB-YJV22-8.7/15-3*240)
Cabinet size (width * depth * height) mm	柜体尺寸(宽*深*高)mm	800*1513*2320	800*1513*2320	800*1513*2320	800*1513*2320	800*1513*2320	800*1513*2320	800*1513*2320	800*1513*2320	800*1513*2320	800*1513*2320
Quadratic map number	二次图图号										

说明:

Description:

1.本图根据深圳市泰昂能源科技有限公司提供的11kV开关柜配电装置接线图绘制。

1.This drawing is based on the wiring diagram of 11kV switchgear distribution device provided by Shenzhen Tai'an Energy Technology Co., Ltd.

2.本工程11kV开关柜包括8面馈线柜，其中I段母线配置3面(M1-3、M1-4、M1-5)，II段母线配置4面(M2-3、M2-4、M2-5、M2-6),另外1面备用(M0-2)配置在工作0段。3面进线柜(M1-1、M0-3、M2-1)，2面分段柜(M0-1、M0-5)，2面隔离柜(M1-6、M2-7),3面母设柜(M1-2、M0-4、M2-2)。

2.The 11kV switchgear of this project includes 8 feeder cabinets, of which the I bus has 3 feeder cabinets(M1-3、M1-4、M1-5), the II bus has 4 feeder cabinets(M2-3、M2-4、M2-5、M2-6), and the other spare one (M0-2) is in the 0 working section. 3 incoming cabinet(M1-1、M0-3、M2-1), 2 sectional cabinets(M0-1、M0-5)，2 isolation cabinets(M1-6、M2-7) and 3 busbar equipment cabinets(M1-2、M0-4、M2-2).

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 2 0 1 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	魏珊	2019.06.07
校核 Checked	郑煜	2019.06.08
审核 Reviewed	张谢平	2019.06.09
批准 Approved	潘益华	2019.06.10

11kV站用电配置接线图

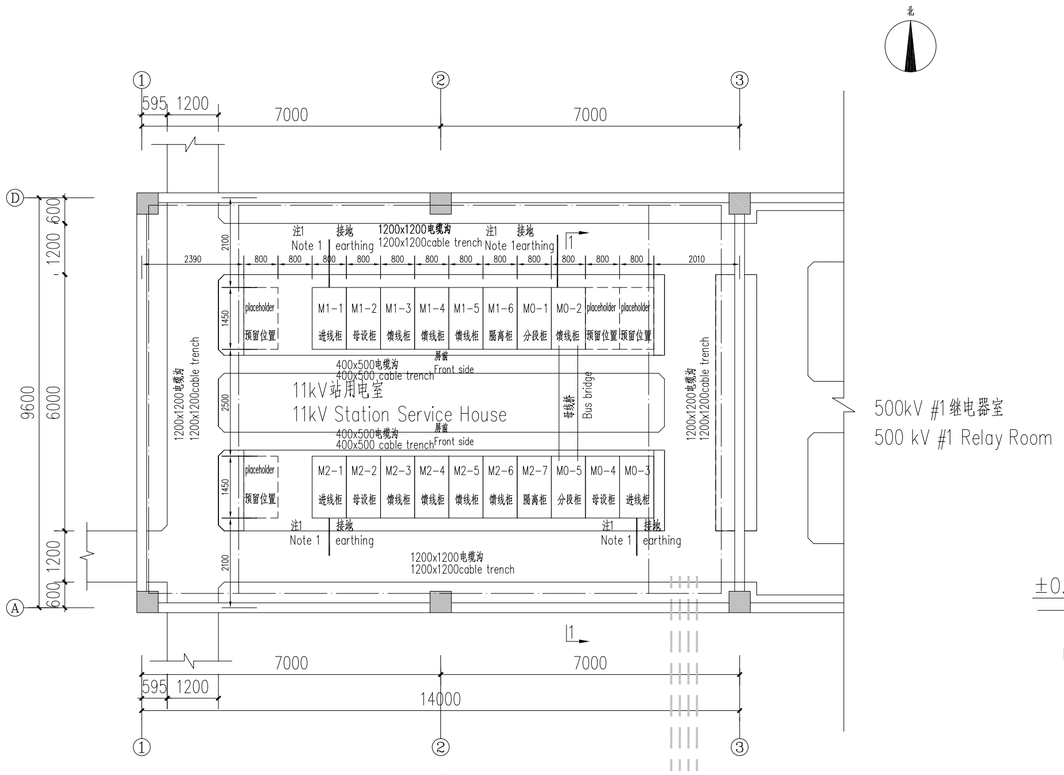
11kV Auxiliary diagram

比例 Scale

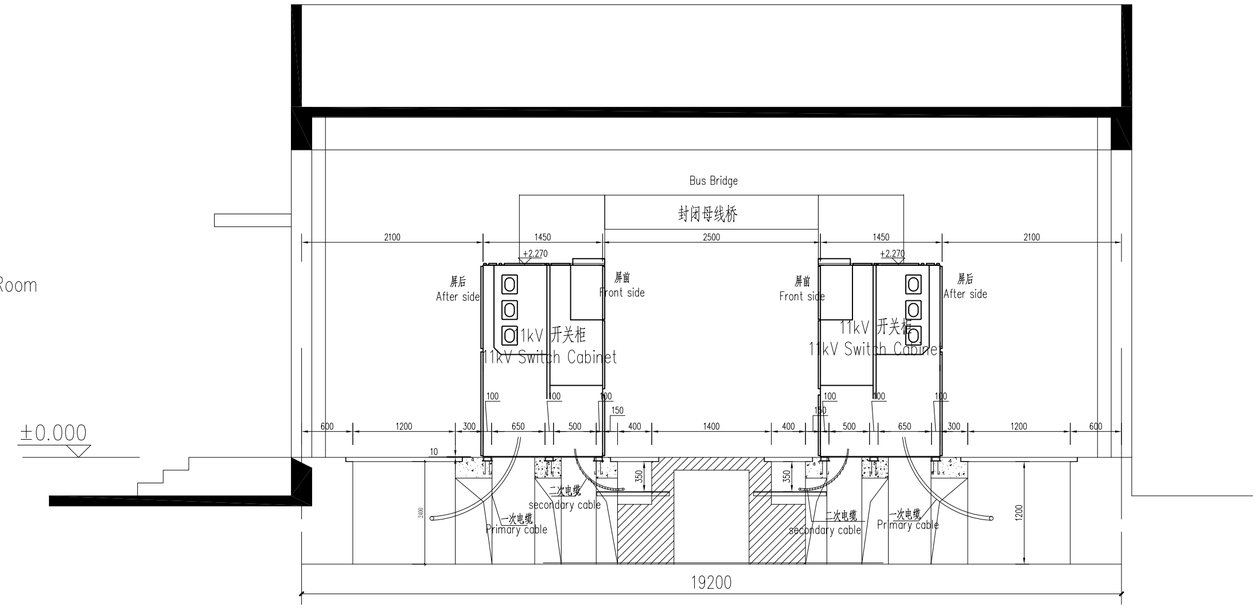
图号 DWG NO. 30-B003701S-D0201-003

版本 REV. A

A2(594x420)



11kV配电室屏柜布置图
11kV cabinet arrangement



1-1断面
1-1section map

注:

Note:

1.11kV开关柜后50*8的接地铜排在M1-1,M0-2,M2-1,M0-3处通过铜绞线(TJ-150)就近与户外主接地网可靠连接。

1.The 50*8 grounding copper bar behind the 11kV switchgear is connected to the ring housing network or outdoor main grounding grid through copper stranded wire (TJ-150) at M1-1, M0-2, M2-1 and M0-3.

2.每面屏柜底部土建已预制3根横向槽钢。屏柜与前后槽钢之间采用点焊方式固定。中间槽钢起支撑作用不需要焊接。

2. Three transverse channel steels have been embedded at the bottom of each screen cabinet.The screen cabinets are fixed by spot welding to front and back channel steel .The intermediate channel provides support and does not require welding.

3. 厂家根据现场实际情况负责提供并安装母线桥吊架。

3.The manufacturer will be responsible for providing and installing bus bridge suspension according to the actual situation on site.

4. 图中标注单位为mm.

4.The unit of marking in the figure is mm.

材料表
material list

序号 Number	名称 Name	规格 Specification	单位 Unit	数量 Quantity	备注 Remark
1	12kV高压开关柜(进线柜) 12kV incoming cabinet	12kV 1250A 25kA	set	3	深圳市泰昂能源科技股份有限公司 Shenzhen Tieon Energy Technology Co.,Ltd
2	12kV高压开关柜(馈线柜) 12kV outlet cabinet	12kV 630A 25kA	set	8	深圳市泰昂能源科技股份有限公司 Shenzhen Tieon Energy Technology Co.,Ltd
3	12kV高压开关柜(分段柜) 12kV sectional cabinet	12kV 1250A 25kA	set	2	深圳市泰昂能源科技股份有限公司 Shenzhen Tieon Energy Technology Co.,Ltd
4	12kV高压开关柜(隔离柜) 12kV isolator cabinet	12kV 1250A 25kA	set	2	深圳市泰昂能源科技股份有限公司 Shenzhen Tieon Energy Technology Co.,Ltd
5	12kV高压开关柜(母设柜) 12kV Busbar Equipment cabinet	/	set	3	深圳市泰昂能源科技股份有限公司 Shenzhen Tieon Energy Technology Co.,Ltd
6	母线桥(含吊架) Bus bridge	12kV 1250A 25kA	m	5	深圳市泰昂能源科技股份有限公司 Shenzhen Tieon Energy Technology Co.,Ltd
7	铜绞线 Copper stranded wire	150mm²	m	40	
8	镀锡铜鼻子 Tinned copper nose	配150mm²铜绞线 Matching 150mm² copper stranded wire	套 set	8	

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程
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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

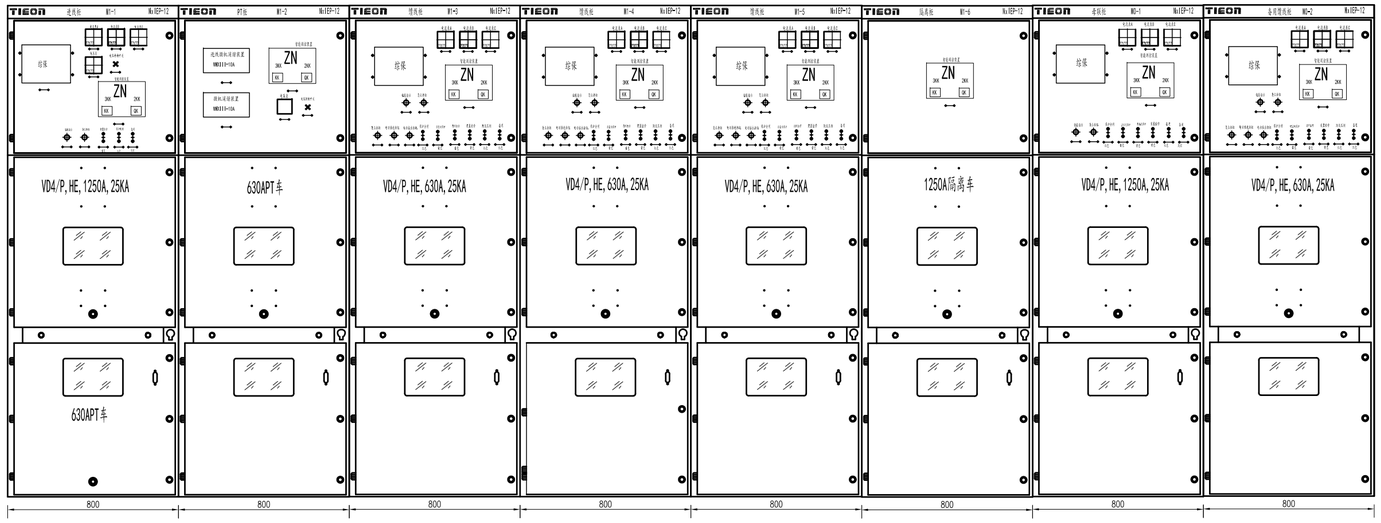
M L H V D C - L C S - E C I - D D - E C - 0 2 0 1 - 0 0 4 A

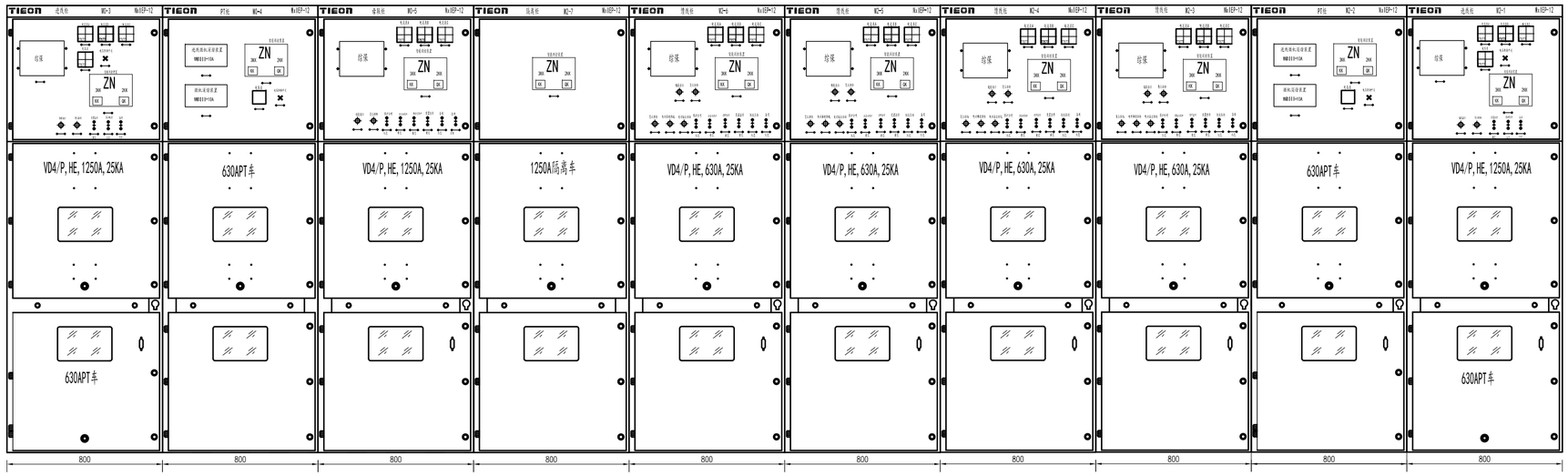
	姓名 Name	日期 Date
设计 Designed	魏珊	2019.06.07
校核 Checked	郑煜	2019.06.08
审核 Reviewed	张谢平	2019.06.09
批准 Approved	潘益华	2019.06.10

11kV站用电室平面布置图
11kV station service power room arrangement

比例 Scale: 1:100 | 图号 DWG NO: 30-B003701S-D0201-004 | 版本 REV: A

A2(594x420)





说明:

Description :

1. 本图按深圳市泰昂能源科技有限公司提供的设备图纸进行绘制。

1. This drawing is drawn according to the equipment drawings provided by Shenzhen Tai'an Energy Technology Co., Ltd.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里–拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

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SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - E C - 0 2 0 1 - 0 0 5 A

	姓名 Name	日期 Date
设计 Designed	魏翀	2019.06.07
校核 Checked	郑煜	2019.06.08
审核 Reviewed	张卿平	2019.06.09
批准 Approved	潘益华	2019.06.10

11kV开关柜正视图

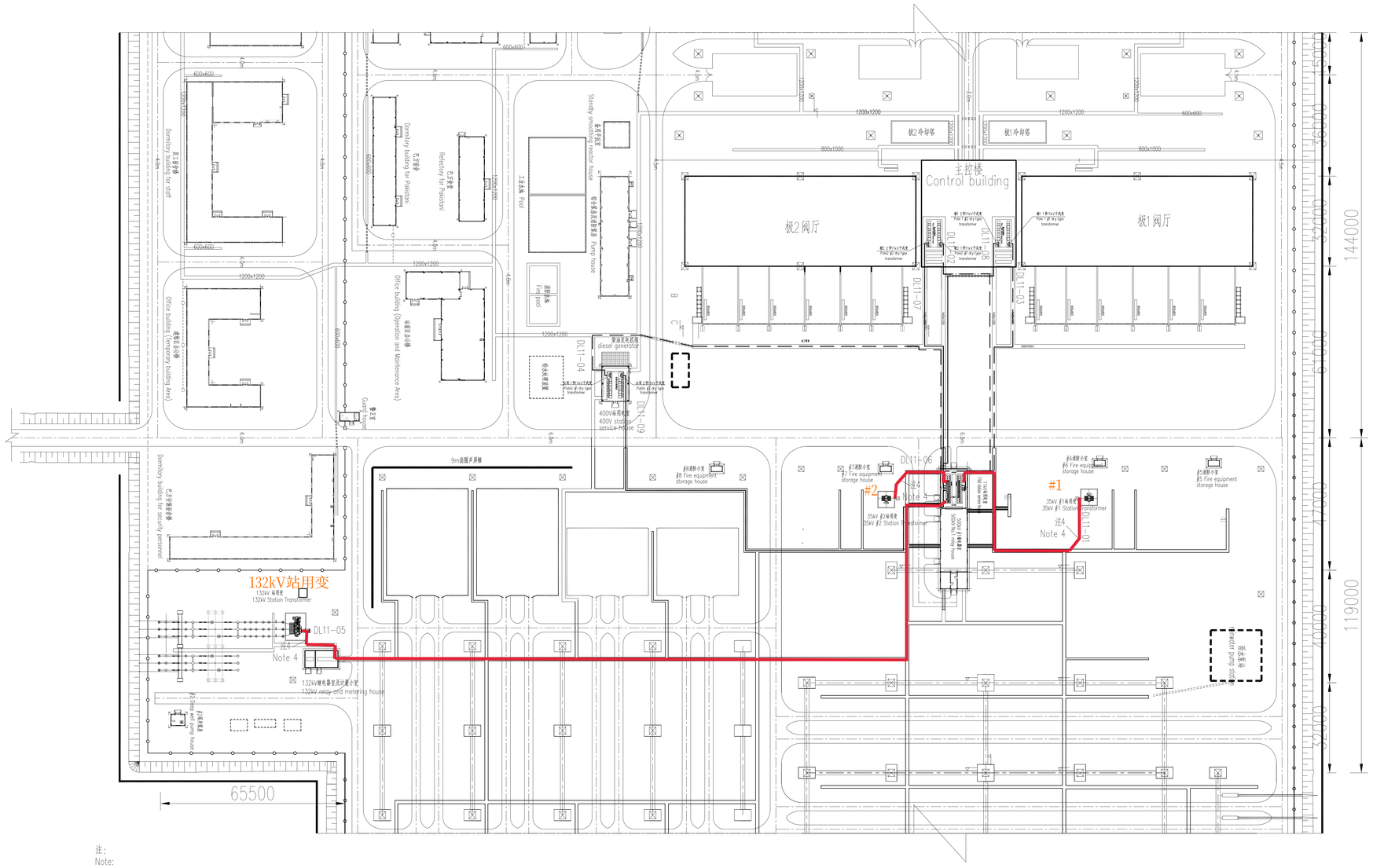
Front view of 11kV cabinets

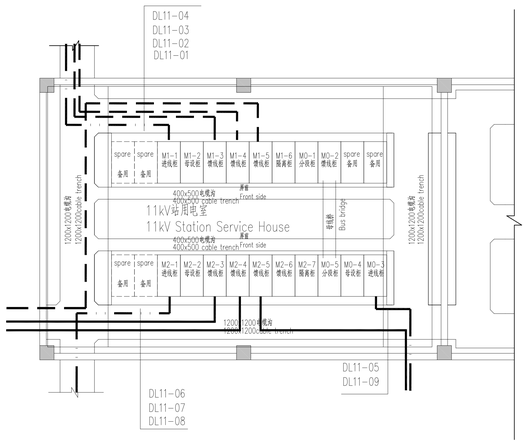
比例 Scale: 1:200

图号 DWG. NO.: 30-B003701S-D0201-005

版本 REV: A

A2(594x420)





11kV配电室屏柜布置图
11kV cabinet arrangement

注:
Note:
1、本站所有11kV电缆布置在电缆沟或埋管内。
1.All 11kV cables in this station are arranged in cable trenches or buried pipes.
2、所有11kV电缆应参照路径图进行敷设，电缆转弯半径一般不得小于15d（d为电缆直径），具体以厂家要求为准。3回11kV进线电源电缆分沟敷设。从不同段11kV母线引接至各动力中心的电缆应尽量分沟敷设，当只能敷设在同一条电缆沟中时，应敷设在不同侧的电缆支架上。
2.All 11kV cables should be laid according to the road map. The turning radius of the cables should not be less than 15d (d is the diameter of the cables). Specific requirements of the manufacturer shall prevail. Three 11 kV power supply cables should be laid in differernt trenches. Cables from different sections of 11kV buses should be laid in differernt trenches if possible. When they can only be laid in the same trench, they should be laid on cable brackets on different sides.
3、应按电压等级由高至低的电力电缆“由下而上”的顺序敷设；其中，最大截面电缆宜敷设在电缆沟最下层支架。11kV 电缆应与400V动力电缆、控制电缆之间采用防火隔板进行分隔。
3.The cable should be laid in the order of "bottom-up" from the highest to the lowest voltage level, and the maximum cross-section cable should be laid on the lowest bracket of the cable trench.11kV cable should be separated from 400V power cable and control cable by fire-proof partition board.
4.为敷设11kV电缆，电气施工单位从35kV #1、#2站用变及132kV站用变处预埋2x⌀200埋管至附近电缆沟。埋管路径应适合图中标识的电缆敷设路径，埋管弯曲半径不小于电缆转弯半径。埋管采用-50x5热镀锌扁钢接地，埋管及热镀锌扁钢开列在本卷册。
4.For laying 11kV cable, electrical construction unit bury 2x⌀200 pipe from 35kV #1、#2 station transformer,132kV station transformer to nearby cable trench.The buried pipe path should be suitable for the cable laying path marked in the figure. The bending radius of the buried pipe should not be less than the bending radius of the cable.Buried pipes are grounded with -50x5 hot-dip galvanized flat steel. Buried pipes and hot-dip galvanized flat steel are listed in this volume.
5、图例
5. Legend:

ZRB-YJV22-8.7/15-3*120 从11kV站用电室至极1 2号、极2 2号、公用2号 11kV干式变
From 11 kV station service house to pole 1 #2, pole 2 #2 and public #2 11 kV dry-type transformer

ZRB-YJV22-8.7/15-3*120 从11kV站用电室至极1 1号、极2 1号、公用1号 11kV干式变
From 11 kV station service house to pole 1 #1, pole 2 #1 and public #1 11 kV dry-type transformer

2*(ZRB-YJV22-8.7/15-3*240) 从11kV站用电室至35kV#1、#2号站用变及132kV站用变11kV侧
From 11 kV station service house to 11kV side of 35 kV#1 and#2 station transformer and 132kV station transformer.

序号 Item No.	电缆编号 Cable No.	型号及规范 Specification	起点 Starting point	终点 End point	单位 Unit	数量 Quanity	备注 Comments
1	DL11-01	2*(ZRB-YJV22-8.7/15-3*240)	35kV #1站用变11kV侧 11 kV Side of 35kV #1 Station Transformer	开关柜M1-1 Switch cabinet M1-1	m	205	已折成单根 Has been converted into single
2	DL11-02	ZRB-YJV22-8.7/15-3*120	开关柜M1-3 Switch cabinet M1-3	极2 1号11kV干式变 Pole2 #1 11kV Dry Transformer	m	128	
3	DL11-03	ZRB-YJV22-8.7/15-3*120	开关柜M1-4 Switch cabinet M1-4	极1 1号11kV干式变 Pole1 #1 11kV Dry Transformer	m	120	
4	DL11-04	ZRB-YJV22-8.7/15-3*120	开关柜M1-5 Switch cabinet M1-5	公用 1号11kV干式变 Public #1 11kV Dry Transformer	m	224	
5	DL11-05	2*(ZRB-YJV22-8.7/15-3*240)	132kV #0站用变11kV侧 11 kV Side of 132kV #0 Station Transforme	开关柜M0-3 Switch cabinet M0-3	m	627	已折成单根 Has been converted into single
6	DL11-06	2*(ZRB-YJV22-8.7/15-3*240)	35kV #2站用变11kV侧 11 kV Side of 35kV #2 Station Transformer	开关柜M2-1 Switch cabinet M2-1	m	77	已折成单根 Has been converted into single
7	DL11-07	ZRB-YJV22-8.7/15-3*120	开关柜M2-3 Switch cabinet M2-3	极2 2号11kV干式变 Pole2 #2 11kV Dry Transformer	m	108	
8	DL11-08	ZRB-YJV22-8.7/15-3*120	开关柜M2-4 Switch cabinet M2-4	极1 2号11kV干式变 Pole1 #2 11kV Dry Transformer	m	121	
9	DL11-09	ZRB-YJV22-8.7/15-3*120	开关柜M2-5 Switch cabinet M2-5	公用 2号11kV干式变 Public #2 11kV Dry Transformer	m	216	
电缆合计 The total cable		2*(ZRB-YJV22-8.7/15-3*240)			m	909	已折成单根 Has been converted into single
		ZRB-YJV22-8.7/15-3*120			m	917	
10	11kV电缆终端 11kV Cable Terminal	配240平方毫米截面的电缆，单相 fit 240 square millimeter cable, single phase			只 set	18	户外冷缩型 Outdoor cold type
11	11kV电缆终端 11kV Cable Terminal	配240平方毫米截面的电缆，单相 fit 240 square millimeter cable, single phase			只 set	18	户内冷缩型 Indoor cold type
12	11kV电缆终端 11kV Cable Terminal	配120平方毫米截面的电缆，单相 fit 120square millimeter cable, single phase			只 set	36	户内冷缩型 Indoor cold type
13	热镀锌钢管 Hot dip galvanized steel pipe	⌀200			m	85	
14	热镀锌扁钢 Hot dip galvanized flat steel	-50x4			m	30	

Reference Drawings

Drwg. No.	Title

Revisions	Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程
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NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 State Grid 中国电力技术装备有限公司 CHINA POWER EQUIPMENT CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H Y D C - L C S - E C I - D D - E C - 0 2 0 1 - 0 0 6 A

	姓名 Name	日期 Date
设计 Designed	魏坤	2019.06.07
校核 Checked	郑煜	2019.06.08
审核 Reviewed	张谢平	2019.06.09
批准 Approved	洪海兵	2019.06.10

11kV电缆敷设路径图
11kV cable laying path diagram

比例 Scale: 1:600 图号 DWG NO.: 30-B003701S-D0201-006 版本 REV: A

841X1189

材料汇总表
Tables of material

序号 Item No.	设备名称 Equipment name	型号及规范 Specification	起点 Starting point	终点 End point	单位 Unit	数量 Quanity	备注 Comments
1	DL11-01	2*(ZRB-YJV22-8.7/15-3*240)	35kV #1站用变11kV侧 11 kV Side of 35kV #1 Station Transformer	开关柜M1-1 Switch cabinet M1-1	m	205	已折成单根 Has been converted into single
2	DL11-02	ZRB-YJV22-8.7/15-3*120	开关柜M1-3 Switch cabinet M1-3	极2 1号11kV干式变 Pole2 #1 11kV Dry Transformer	m	128	
3	DL11-03	ZRB-YJV22-8.7/15-3*120	开关柜M1-4 Switch cabinet M1-4	极1 1号11kV干式变 Pole1 #1 11kV Dry Transformer	m	120	
4	DL11-04	ZRB-YJV22-8.7/15-3*120	开关柜M1-5 Switch cabinet M1-5	公用 1号11kV干式变 Public #1 11kV Dry Transformer	m	224	
5	DL11-05	2*(ZRB-YJV22-8.7/15-3*240)	132kV #0站用变11kV侧 11 kV Side of 132kV Station Transformer	开关柜M0-3 Switch cabinet M0-3	m	627	已折成单根 Has been converted into single
6	DL11-06	2*(ZRB-YJV22-8.7/15-3*240)	35kV #2站用变11kV侧 11 kV Side of 35kV #2 Station Transformer	开关柜M2-1 Switch cabinet M2-1	m	77	已折成单根 Has been converted into single
7	DL11-07	ZRB-YJV22-8.7/15-3*120	开关柜M2-3 Switch cabinet M2-3	极2 2号11kV干式变 Pole2 #2 11kV Dry Transformer	m	108	
8	DL11-08	ZRB-YJV22-8.7/15-3*120	开关柜M2-4 Switch cabinet M2-4	极1 2号11kV干式变 Pole1 #2 11kV Dry Transformer	m	121	
9	DL11-09	ZRB-YJV22-8.7/15-3*120	开关柜M2-5 Switch cabinet M2-5	公用 2号11kV干式变 Public #2 11kV Dry Transformer	m	216	
电缆合计 The total cable		2*(ZRB-YJV22-8.7/15-3*240)			m	909	已折成单根 Has been converted into single
		ZRB-YJV22-8.7/15-3*120			m	917	
10	12kV高压开关柜 12kV high voltage switch cabinet	[深圳泰昂]KYN28A-12 800*1513*2320(宽*深*高) [Shenzhen Tieon]KYN28A-12 800*1513*2320(Width * depth * height)			面 set	18	甲供 provided by party A
11	11kV电缆终端 11kV Cable Terminal	配240平方毫米截面的电缆，单相 fit 240 square millimeter cable, single phase			只 set	18	户外冷缩型 Outdoor cold type
12	11kV电缆终端 11kV Cable Terminal	配240平方毫米截面的电缆，单相 fit 240 square millimeter cable, single phase			只 set	18	户内冷缩型 Indoor cold type
13	11kV电缆终端 11kV Cable Terminal	配120平方毫米截面的电缆，单相 fit 120 square millimeter cable, single phase			只 set	36	户内冷缩型 Indoor cold type
14	11kV封闭母线桥 11kV Bus bridge	1250A 25kA ,三相共箱 1250A 25kA ,three-phase box			m	5	深圳泰昂配套提供，施工单位安装 Shenzhen Tieon provide ,Construction unit installe
15	铜绞线 Copper stranded wire	150mm²			m	40	
16	镀锡铜鼻子 Tinned copper nose	配150mm²铜绞线 Matching 150mm² copper stranded wire			套 set	8	
17	热镀锌钢管 Hot dip galvanized steel pipe	ø200			m	85	
18	热镀锌扁钢 Hot dip galvanized flat steel	-50x4			m	30	

说明:
Description:
1. 除特殊说明外，表中材料均由电气施工单位提供。
1. Except for special instructions, all materials in the table are provided by the electrical construction unit.

Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

MLHVDC-LCS-ECI-DD-EC-0201-007A

	姓名 Name	日期 Date
设计 Designed	魏聊	2019.06.07
校核 Checked	郑煜	2019.06.08
审核 Reviewed	张谢平	2019.06.09
批准 Approved	潘益午	2019.06.10

材料汇总表
Material summary list

比例 Scale | 图号 DWG NO. 30-B003701S-D0201-007 | 版本 REV A

297X420